Srihari Vemuru

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Skills

Languages: C/C++ (Codeforces: Specialist), Embedded C++, Python, Rust, Java, JavaScript

Robotics Software: ROS2, Gazebo, Isaac Sim, MuJoCo, CARLA, RViz, PX4 Autopilot, MavLink

Perception & ML: OpenCV, Open3D, ORB-SLAM3, COLMAP, YOLOv11, PyTorch, TensorFlow, vLLM

Embedded & Systems: Embedded Linux, NVIDIA Jetson, Raspberry Pi 4/5, Pixhawk, Linux/Unix, Multi-threading/Concurrency, CMake, Docker, Git

Experience

Incoming Research Intern Jan 2027 – Apr 2027
Mitsubishi Electric Research Labs (MERL) Cambridge, MA

- Building onboard **mapping software** and **fleet task allocation** for a heterogeneous aerial + ground robot fleet.

Research Assistant Aug 2024 – Aug 2026
Iowa State University Ames, IA, USA

- Built a **Vicon + Crazyflie** testbed proving **sim-to-real transfer** on physical hardware, validating the full pipeline end to end from autonomous takeoff and hover to point-to-point motion and landing.
- Developed a **Python SDK bridge** streaming Vicon 6-DOF pose data into the Crazyflie's **onboard Kalman estimator**.
- Developed a **real-time YOLOv11** detection pipeline on-board **NVIDIA Jetson** for autonomous delivery robots.

Robotics Engineer Sept 2022 – July 2024
Strider Robotics Bengaluru, India

- ASME IMECE 2026, ICRA 2027 (under review): Developed **3D SLAM-based multi-object trackers** beating SOTA by up to 30% on three agricultural datasets, and by 98% online using **probabilistic 3D landmarks** for scale-invariant re-ID.
- Built a **C++/ROS2** quadruped navigation stack (A*, Gazebo), collision-free in dynamic **outdoor environments**.
- Fused GPS and IMU with an **EKF** to synchronize **real-time localization** across a fleet of 2 UAVs and a Go1 quadruped.
- Deployed **vLLM** across an 8-GPU server as an internal OpenAI-compatible API, and wrote a Python wrapper over the Vast.ai API to launch, monitor and auto-terminate GPU instances, cutting **idle-GPU spend** by 30%.

Education

Iowa State University Aug 2024 – Dec 2026
Masters in Mechanical Engineering | GPA: 3.67 Ames, IA

IIIT Bangalore Aug 2017 – July 2022
Masters and Bachelors in Computer Science | GPA: 3.42 Bangalore, India

Projects

SLAM Toolbox | Open Source Contribution | C++, ROS2

- Fixed an RViz **TF frame mismatch** and added **max-range ray clearing** to the C++ library and ROS2 nodes.
- Added **runtime switching** between mapping and localization without restarting the launch stack.

Agricultural Field Robotics Platform | Strider Robotics | C++, ROS2, SolidWorks

- Built a **C++/ROS2** navigation and planning stack for a quadruped-arm **precision-agriculture** sprayer.
- Designed a CAD-modelled, **3D-printed sliding dock** for UAV-assisted, in-field **battery swapping**.

LeRobot | Open Source Contribution | Python, PyTorch

- Made **SmolVLA**'s from-scratch path **backbone-agnostic**, replacing hard-coded SmolVLM class and attribute lookups so the VLM backbone can be swapped.
- Fixed a **horizontal flip** of LIBERO image observations and a **float32 precision** error in video timestamp concatenation.